Biochemistry Cume September 10, 2010

When protein amino acid sequences are aligned, amino acid residues in the respective sequences are categorized as 'identical' or 'mismatched.' Sometimes a third, albeit subjective, category is used to indicate 'conservative' changes or 'similar' amino acids. In the aligned sequences below, such 'conservative' sequence modifications are denoted by + symbols in the comparison line between the two different sequences.

The main purpose of this cume is to evaluate the assignments of 'conservative' or 'positive' changes. A table to compare the one-letter and 3-letter amino acid codes is given at the end.

Alignment of two arbitrary protein sequences:

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Identities = 60/190 (31%), Positives = 100/190 (52%), Gaps = 6/190 (3%),
Frame = -3

Query  179      SIVNAFMVGCMFVKISQPKKRAETLVFSTHAVISMRDGKLCLMFRVGDLRNSHIVEASIR
                +I   F +G    KI++PK RA ++ F+  AV++ +DGK  L+F+V + R S +    +
Sbjct  1761381  NISMVFNLGAFVAKIARPKNRAFSIRFTDLAVVAHKDGKPNLIFQVANTRPSPLTNVRVS

Query  239      AKLIKSKQTSEGEFIPLNQTDINVGYYTGDDRLFLVSPLIISHEINQQSPFWEISKAQLP
                A L + ++  E   +     D ++   + ++  F + PL   H I+  SP   + + + P
Sbjct  1761201  AVLYQERENGE---LYQTSVDFHLDGISSEECPFFIFPLTYYHTISPSSPLATLLQHETP

Query  299      KEELEIVVILEGMVEATGMTCQARSSYITSEILWGYRFTPVLTL-EDGFYEVDYNSFHET
                    E+VV L  M E TG  CQ R+SY+ SEI+  +RF  ++T    G Y+V   +F +T
Sbjct  1761030  -PHFELVVFLSAMQEGTGEICQRRTSYLPSEIMLHHRFAALMTRGSKGEYQVKMENFDKT
```

Questions:

1. (68 points) Seventeen different claimed amino acid side-chain similarities are marked (+) in the above alignment. For each different pair that is marked (+), state a key similarity **AND** a key difference between the pair of amino acid side chains.
For your answers, please use a **table format** similar to the example shown below.
(We may not be able to grade answers that are not organized into a table.)

example Table:

Residue Pair		Key Similarity	Key Difference
S	N		
V	L		
S	A		

(continue)

(continued)

good and poor choices:

2. (10 points) Nominate two pairs from the list of 17 that are among the most closely similar to each other. Justify your choices as being among the best choices in the list.

3. (8 points) Nominate two pairs from the list of 17 that are among the least closely similar to each other. Justify your choices as being among the weakest or least defensible choices in the list.

other good choices, not in the above set of 17 pairs:

4. (8 points) Nominate two pairs of closely similar amino acids that are not included in the list of 17 from the example alignment. Justify your choices as to why these pairs are similar to each other.

final question:

5. (6 points) In the alignment example, explain the meaning of the phrase "frame = -3."

Appendix

Comparison of the one-letter and three-letter amino acid codes.

A	Ala
C	Cys
D	Asp
E	Glu
F	Phe
G	Gly
H	His
I	Ile
K	Lys
L	Leu
M	Met
N	Asn
P	Pro
Q	Gln
R	Arg
S	Ser
T	Thr
V	Val
W	Trp
Y	Tyr